A survey on Reinforcement Learning from Human Feedback for Large Language Models

Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in various natural language processing (NLP) tasks, such as machine translation, text generation, and question answering. However, these models often fail to generate human-aligned outputs, exhibiting biases, incoherency, and lack of diversity in their responses. One promising approach to overcome these limitations is Reinforcement Learning from Human Feedback (RLHF) , which enables LLMs to iteratively improve their outputs based on human evaluative feedback. This technique leverages reinforcement learning (RL) algorithms to fine-tune model behavior by using human feedback as a reward signal. RLHF is seen as a powerful method for aligning LLMs with human values and improving their generation quality across a range of tasks.

In this paper, we provide a comprehensive overview of the RLHF technique in the context of LLMs. We explore the background and foundational principles of RLHF, discuss the main challenges and limitations it faces, and examine the state-of-the-art methods developed to address these challenges. Additionally, we consider future research directions and potential applications of RLHF in training more responsible and human-aligned LLMs.

1 Introduction

Large Language Models (LLMs)have achieved impressive performance across a wide array of natural language processing (NLP) tasks, from generating coherent and contextually relevant text to performing complex reasoning

tasks.[2, 4] Models such as GPT, OPT, and Llama [1, 22, 20]have demonstrated state-of-the-art results, leveraging vast amounts of unsupervised data to learn language patterns. Despite their success, these models frequently generate outputs that do not align with user intent or societal values. They may produce biased content, irrelevant information, or even offensive text, posing significant challenges when deployed in real-world applications.

Traditional training methods for LLMs, primarily based on supervised learning, depend on vast amounts of labeled data[23]. However, this approach fails to fully capture the nuances of human expectations. Even when fine-tuned on domain-specific datasets, LLMs can still misinterpret complex or subjective user inputs, leading to suboptimal or harmful outputs. Moreover, supervised learning methods do not allow for continuous model improvement once the training phase is complete.

Reinforcement Learning from Human Feedback (RLHF)[16] has emerged as a promising solution to this problem, enabling models to learn from human evaluators instead of relying solely on pre-labeled data. By incorporating human feedback as a reward signal in the reinforcement learning process, RLHF allows LLMs to adjust their behavior in an iterative fashion, making them better aligned with human goals, preferences, and ethical standards. RLHF leverages the ability of reinforcement learning to optimize complex decision-making tasks by aligning model outputs with human feedback.

This review explores RLHF in the context of LLMs, delving into the background, limitations, challenges, and methods associated with this technique. We also discuss the ongoing research efforts and potential future directions for RLHF in making LLMs more adaptable, responsible, and human-centric.

2 Background

2.1 The misalignment of Large Language Models

Large Language Models (LLMs) have ushered in a transformative era in Natural Language Processing (NLP), empowering machines to learn from vast, diverse textual data and to capture the intricate linguistic structures and deep semantic relationships that underpin human language. By leveraging sophisticated unsupervised and semi-supervised learning techniques, these models excel in generating human-like text, anticipating the next word in a sequence based on the provided context. Despite these remarkable ad-

vancements, however, LLMs often face significant challenges when it comes to aligning with human values, ethical standards, and practical needs[9].

A fundamental issue lies in the fact that LLMs are typically trained on large, static datasets, which, while expansive, are inherently flawed. These datasets may harbor biases, inaccuracies, and imbalances that influence the model's outputs in undesirable ways[6]. For instance, models trained on unfiltered internet data may inadvertently perpetuate harmful stereotypes, spread misinformation, or reinforce societal inequalities[8]. Beyond data bias, LLMs also struggle with the subtleties of human intent and emotional context. This is particularly evident in situations requiring empathy, nuance, or ethical judgment, such as negotiations, customer support, or content moderation. In these scenarios, the misalignment between the model's predictions and the user's true intent or ethical standards can lead to problematic, and even damaging, results.

Moreover, the challenges of misalignment are compounded by the black-box nature of these models. Their decision-making processes are often opaque, making it difficult for users to understand why a particular output is generated, let alone ensure that it aligns with ethical guidelines or societal norms[8]. As LLMs are increasingly deployed in critical applications that directly impact individuals and society, addressing this misalignment has become a pressing concern, demanding both technical solutions and careful consideration of broader ethical implications.

2.2 Reinforcement Learning from Human Feedback

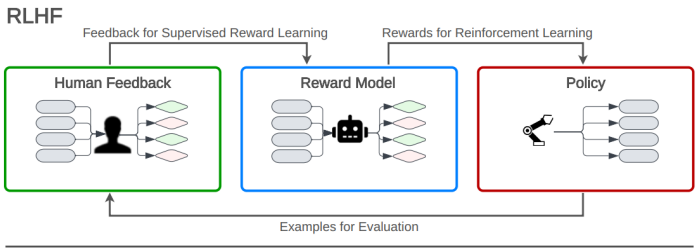


Figure 1: Diagram of the RLHF model[3]

Reinforcement Learning from Human Feedback (RLHF) is a machine learning technique that uses human feedback to train a "reward model,"

which in turn optimizes the performance of AI agents through reinforcement learning. This method is particularly valuable for tasks with complex or ill-defined goals, such as generating humor in text. RLHF was introduced in a 2017 paper by researchers from OpenAI and DeepMind, who demonstrated its success in training AI models for tasks like playing Atari games and simulating robotic locomotion. The introduction of the Proximal Policy Optimization (PPO) algorithm significantly reduced the cost of gathering human feedback, facilitating the integration of RLHF with natural language processing (NLP). OpenAI's release of InstructGPT in 2022, trained using RLHF, was a pivotal step in advancing large language models (LLMs)[5].

In RLHF, two models are trained: a reward model that learns desirable behaviors from human feedback, and a reinforcement learning policy that guides the agent's actions based on this reward model. Both models are typically initialized with pre-trained autoregressive language models and further trained on datasets of human-annotated prompt-response pairs. The reward model is trained by replacing the final layer of the pre-trained model with a regression head, shifting the model from classification to outputting a score for any given prompt and response[7]. The model is trained on human preference comparison data, minimizing the cross-entropy loss function to make predictions closer to actual human ratings.

RLHF has been applied across various NLP domains, including conversational agents, text summarization, and natural language understanding[16]. It has also been used to align text-to-image models in computer vision. The use of KL regularization in RLHF helps stabilize the training process by preventing the learned policy from deviating too far from the initial model, thus reducing overfitting to the reward model.

3 Limitations and Challenges of RLHF

Reinforcement Learning from Human Feedback (RLHF) has emerged as a pivotal technique in training AI agents, particularly in the domain of natural language processing (NLP). However, despite its significant advancements, RLHF faces several limitations and challenges that need to be addressed for more effective and broader application.

Scalability and Resource Intensity. Reinforcement Learning with Human Feedback (RLHF) presents significant scalability challenges due to its reliance on extensive human and monetary resources[14]. Collecting mean-

ingful and actionable feedback from human evaluators is a labor-intensive process that often requires training annotators, designing clear evaluation guidelines, and ensuring consistency across feedback. Additionally, the cost associated with compensating human contributors and maintaining infrastructure for large-scale data collection can be prohibitive. This resource-intensive nature makes RLHF less accessible to smaller organizations and limits its scalability to broader applications.

Quality and Reliability of Human Feedback. The effectiveness of RLHF hinges on the quality of human feedback, which is often prone to variability and noise. Human evaluators may interpret tasks differently, introduce subjective biases, or provide inconsistent assessments based on factors such as fatigue, lack of domain expertise, or unclear instructions. Poorly curated or low-quality feedback can misguide the reward model, leading to suboptimal policy updates and degraded model performance. Addressing these challenges requires robust mechanisms to filter, aggregate, and validate feedback, as well as clear task definitions to minimize ambiguity. [10]

Alignment with Human Values. Ensuring that AI systems align with human values is a deeply complex and multifaceted challenge[15] . Human values are diverse, context-dependent, and often subject to cultural or individual differences, making it difficult to define universal standards for alignment. RLHF seeks to approximate alignment by modeling human preferences, but this process is inherently imprecise and may fail to capture nuanced ethical considerations. Furthermore, unintended biases in human feedback can inadvertently propagate into the model, potentially leading to behaviors that conflict with intended values. Achieving meaningful alignment requires iterative refinement, diverse feedback sources, and ongoing monitoring of model outputs.

Technical and Algorithmic Limitations. The RLHF pipeline introduces significant technical and algorithmic challenges, as it integrates multiple sophisticated processes. The initial step—gathering human feedback—requires careful design to ensure evaluators provide actionable and unbiased input. Reward modeling, the second step, involves creating a model that accurately reflects human preferences, which is non-trivial due to the complexity of translating subjective feedback into a quantifiable reward signal. Finally, the policy optimization step requires advanced reinforcement learning techniques to train the model effectively while balancing exploration and exploitation. Each stage demands expertise in machine learning, careful system design, and extensive computational resources. [18]

Failure Modes and Robustness. Despite its potential, RLHF-trained models are susceptible to various failure modes, which can undermine their reliability and safety[17]. These failure modes may manifest as unintended behaviors, reward hacking (where the model exploits loopholes in the reward system), or poor performance in scenarios not covered by the training feedback. Additionally, models trained with RLHF may lack robustness when exposed to adversarial inputs or unforeseen edge cases, highlighting the need for thorough testing and evaluation. Addressing these issues requires robust validation protocols, diversified training datasets, and ongoing refinement of the reward system.

Generalization and Overfitting. One of the critical challenges in RLHF is achieving strong generalization while avoiding overfitting to the training data or reward model[11]. Overfitting occurs when the model becomes excessively attuned to the specific patterns or preferences captured in the training feedback, resulting in degraded performance in novel or real-world scenarios. This problem is exacerbated by the often limited diversity of human-provided feedback, which may not encompass the full range of potential use cases. Strategies to improve generalization include increasing the diversity of training environments, regularizing the reward model, and adopting techniques such as domain randomization or transfer learning.

While RLHF has shown significant promise in training AI agents, particularly in NLP tasks, it faces several limitations and challenges. Addressing these challenges through innovative methods and techniques, as proposed in the recent literature, is crucial for the broader and more effective application of RLHF in real-world scenarios.

4 Methods

To overcome the challenges of RLHF, several methods and techniques have been developed to enhance the effectiveness and efficiency of the feedback loop.

4.1 RLAIF: Scaling Reinforcement Learning from Human Feedback with AI Feedback

The paper *RLAIF: Scaling Reinforcement Learning from Human Feedback with AI Feedback* [14] introduces a method to scale reinforcement learning by

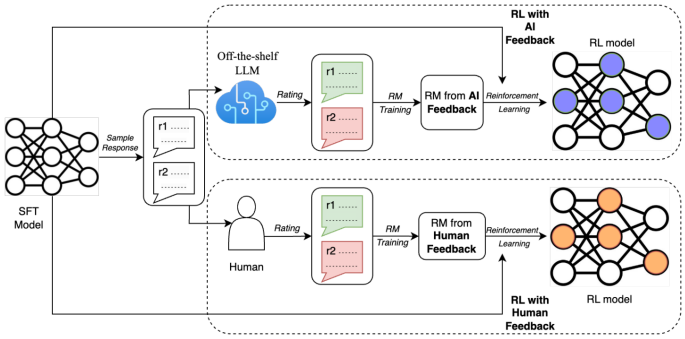


Figure 2: A diagram depicting RLAIF (top) vs. RLHF (bottom)[14]

utilizing AI-generated feedback as an alternative to human annotations. The approach leverages an off-the-shelf large language model (LLM) to generate preference labels, which are then used to train a reward model (RM) for reinforcement learning (RL). This innovation addresses the scalability challenges of Reinforcement Learning from Human Feedback (RLHF), which depends on costly and time-intensive human annotations.

Methodology

I. AI Feedback Generation: Given an input x, a pair of responses (y_1, y_2) is sampled from one or more models, often starting with a supervised fine-tuned (SFT) model. The input and responses are then sent to a pre-trained LLM, which generates preference labels by evaluating which response better meets predefined criteria. The preference distribution is calculated as:

$$P(i \mid y, x) = \frac{\exp(\log P(i \mid y, x))}{\sum_{j=1}^{2} \exp(\log P(j \mid y, x))}, \tag{1}$$

where $P(i \mid y, x)$ represents the probability of response i being preferred. The response with the higher probability is selected as the preferred one.

II. Reward Model Training: A reward model r_ϕ is trained using the AI-generated preference labels. The loss function for training is defined as:

$$L_r(\phi) = -E_{(x,y_w,y_l)\sim D} \left[\log \sigma \left(r_\phi(x, y_w) - r_\phi(x, y_l)\right)\right], \tag{2}$$

where σ is the sigmoid function, D is the dataset of triplets (x, y_w, y_l), and y_w, y_l are the preferred and non-preferred responses, respectively.

III. Reinforcement Learning: The policy model π_{RL}^{θ} is initialized from the SFT model and optimized using reinforcement learning to maximize rewards provided by the RM. To prevent over-optimization or unnatural outputs, an optional Kullback-Leibler (KL) divergence loss is added:

$$\mathcal{L}(\theta) = E_{(x,y)\sim\pi_{\text{RL}}^{\theta}}\left[r_{\phi}(x,y)\right] - \beta D_{\text{KL}}\left(\pi_{\text{RL}}^{\theta} \parallel \pi_{\text{SFT}}\right), \tag{3}$$

where β is a regularization coefficient, and D_{KL} penalizes deviations from the SFT model π_{SFT}.

IV. Direct-RLAIF (d-RLAIF): The d-RLAIF variant bypasses reward model training by directly using LLM feedback as a reward signal during RL. The LLM assigns a quality score $s(y \mid x)$ to a response y, computed as:

$$s(y \mid x) = \sum_{i=1}^{10} i \cdot P(i \mid y, x), \tag{4}$$

where scores are normalized to the range $[-1, 1]$ for reinforcement learning optimization.

Results

Summarization Task: RLAIF and RLHF outperform the supervised fine-tuned baseline in human evaluations, with each method being preferred approximately 70% of the time. When compared directly, human preferences for RLAIF and RLHF are statistically indistinguishable, demonstrating comparable performance.

Helpful Dialogue Generation: In dialogue tasks, RLAIF and RLHF exceed the baseline's performance, with human evaluators preferring these methods 63% and 64% of the time, respectively. Notably, d-RLAIF matches or surpasses traditional RLAIF, validating the direct usage of LLM feedback.

Harmless Dialogue Generation: Evaluations of response harmlessness reveal that RLAIF achieves an 88% harmlessness rate, surpassing RLHF (76%) and the baseline (64%). This result underscores the potential of RLAIF to generate safer outputs while maintaining alignment with human preferences.

Conclusion

The RLAIF framework demonstrates that AI-generated feedback can scale reinforcement learning effectively, matching the performance of RLHF while reducing dependence on costly human annotations. Future work may explore hybrid approaches that combine human and AI feedback to further enhance performance and robustness.

4.2 Smaug: Fixing Failure Modes of Preference Optimization with DPO-Positive

The paper *"Smaug: Fixing Failure Modes of Preference Optimization with DPO-Positive"* [17]introduces a novel method called DPO-Positive (DPOP) to address the failure modes of Direct Preference Optimization (DPO) in preference-based fine-tuning. The authors demonstrate that DPOP effectively mitigates these failure modes and achieves state-of-the-art results on various benchmarks.

Methodology

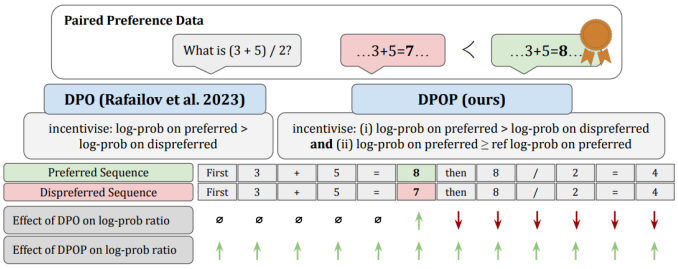


Figure 3: DPOP avoids a failure mode of DPO[17]

I. Failure Modes in DPO

- The paper highlights a critical issue where applying DPO to preference datasets with minimal differences (low edit distances) between completions can significantly degrade performance.
- Theoretical analysis reveals that DPO's gradient updates may reduce the log-likelihood of preferred examples, counteracting its intended purpose.

II. DPO-Positive (DPOP) Loss Function DPOP is designed to ensure that the log-likelihood of preferred examples remains high relative to the reference model. The DPOP loss function is defined as:

$$L_{\text{DPOP}}(\pi_\theta; \pi_{\text{ref}}) = - E_{(x,y_w,y_l)\sim D}\left[\log \sigma\left(\beta\left(\log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right)\right] - \lambda \cdot \max\left(0, \log \frac{\pi_{\text{ref}}(y_w \mid x)}{\pi_\theta(y_w \mid x)}\right)$$

where $\lambda > 0$ is a hyperparameter that penalizes significant reductions in the likelihood of preferred examples. This formulation prevents excessive reductions in the log-likelihood of dispreferred examples by simultaneously optimizing the relative likelihoods of preferred examples compared to the reference model.

III. Dataset Construction The authors create preference-ranked versions of widely used datasets, including ARC, HellaSwag, and MetaMath, to validate DPOP. For example, a preference-annotated MetaMath dataset is developed, extending GSM8K training data for fine-tuning.

Results

Performance on MetaMath

- On the MetaMath dataset, characterized by low edit distances, DPO and other methods such as SLiC fail to improve performance, often degrading accuracy.
- DPOP, however, significantly improves model performance, demonstrating its robustness in challenging scenarios.

Performance on ARC

- On the ARC dataset, which features higher edit distances, DPOP outperforms other methods, including DPO and SLiC, achieving superior fine-tuning outcomes.

HuggingFace Open LLM Leaderboard

- **Smaug-72B**, a model fine-tuned with DPOP, achieves an average accuracy of 80.48%, becoming the first open-source large language model (LLM) to exceed this threshold. This represents a nearly 2% improvement over the second-best open-source model.
- **Smaug-34B**, another DPOP-trained model, emerges as the best-performing model in its parameter class, further validating the scalability and effectiveness of DPOP.

Conclusion

The introduction of DPO-Positive (DPOP) resolves critical limitations of DPO, enabling consistent and robust improvements across diverse datasets. By fine-tuning with DPOP on new preference-ranked versions of ARC, HellaSwag, and MetaMath, the Smaug models set new benchmarks in LLM performance, with Smaug-72B achieving record-breaking accuracy on the HuggingFace Open LLM Leaderboard. These results highlight DPOP's potential to advance preference-based optimization in large language models.

4.3 Reference-free Monolithic Preference Optimization with Odds Ratio

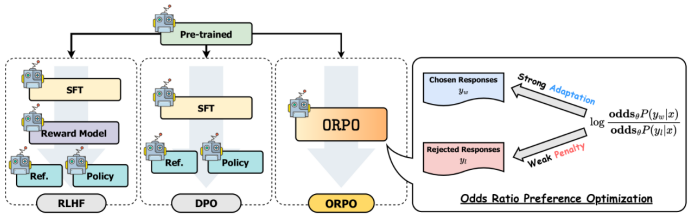


Figure 4: Comparison of model alignment techniques. ORPO aligns the language model without a reference model in a single-step manner by assigning a weak penalty to the rejected responses and a strong adaptation signal to the chosen responses with a simple log odds ratio term appended to the negative log-likelihood loss.[11]

The paper *"Reference-free Monolithic Preference Optimization with Odds Ratio"* [11]proposes a novel method called Odds Ratio Preference Optimization (ORPO) to align language models with human preferences without the need for a reference model. This method aims to enhance the efficiency and performance of preference alignment by eliminating the necessity for an additional alignment phase, thereby reducing compute and memory requirements.

Methodology

I. Odds Ratio Preference Optimization (ORPO)

- ORPO introduces a reference-free preference optimization framework that integrates preference alignment directly into the supervised fine-tuning (SFT) process.
- The core idea is to use the odds ratio to penalize undesired generation styles while reinforcing preferred styles. This simplifies the optimization process and avoids reliance on a separate reference model.

II. Loss Function The ORPO loss function modifies the negative log-likelihood (NLL) by incorporating a log odds ratio term to contrast favored (y_w) and disfavored (y_l) responses:

$$L_{\text{ORPO}}(\theta) = -E_{(x,y_w,y_l)\sim D}\left[\log \sigma\left(\beta\left(\log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)}\right)\right)\right]$$

where σ is the sigmoid function, β is a hyperparameter, and π_θ is the model being trained. Note that ORPO does not utilize a reference model, making the process monolithic and resource-efficient.

III. Training Process

- ORPO integrates directly into the SFT phase, optimizing the model on a dataset of pairwise preferences.
- The method penalizes the generation of disfavored responses while rewarding favored ones, eliminating the need for a separate warm-up phase or external reference model.

Results

Model Performance

- ORPO was tested on Phi-2 (2.7B), Llama-2 (7B), and Mistral (7B) models using the UltraFeedback dataset.
- Mistral-ORPO-β (7B) achieved:
  - 7.32 in MT-Bench,
  - 91.41% in AlpacaEval 1.0,
  - 12.20% in AlpacaEval 2.0,

  outperforming state-of-the-art models such as Zephyr-β (7B), TULU-2-DPO (13B), and Llama-2-Chat (13B).

Empirical and Theoretical Justification

- Empirical results and theoretical analysis validate the use of the odds ratio for contrasting favored and disfavored styles during SFT.
- ORPO demonstrates consistent performance improvements across models of varying sizes (from 125M to 7B parameters).

Resource Efficiency

- ORPO eliminates the need for an additional preference alignment phase, reducing computational overhead.
- This makes the method particularly advantageous for large-scale language models where resource efficiency is critical.

Conclusion

The paper introduces ORPO as an efficient, reference-free preference optimization framework that effectively aligns language models with human preferences. By leveraging the odds ratio during supervised fine-tuning, ORPO achieves state-of-the-art performance on multiple benchmarks, while significantly reducing resource requirements. This approach holds promise for advancing preference alignment in natural language processing.

5 Discussion

In recent years, Reinforcement Learning from Human Feedback (RLHF) has emerged as a powerful technique for aligning artificial intelligence (AI) systems with human values and preferences. By leveraging human feedback, RLHF enables AI models, particularly large language models (LLMs), to adapt and improve their performance in ways that reflect human goals. However, while RLHF shows great promise in addressing alignment challenges, it also presents significant technical and sociotechnical issues. The primary limitation of RLHF lies in its ability to address the complexity of human values comprehensively, as AI alignment involves not only technical solutions but also societal considerations. Despite its successes, RLHF continues to face challenges in ensuring fairness, avoiding biases, and maintaining transparency. Moreover, as noted in [5, 12], unresolved issues such as reward modeling, designing recursive mechanisms, and aligning models' behavior with complex and diverse societal values highlight the need for further research and refinement.

Incorporating RLHF into LLMs offers the potential to improve their behavior, particularly by making them more responsive to human input. However, the technique's reliance on human feedback requires careful attention to the quality and diversity of the feedback. Labeler biases and the limited representativeness of feedback sources can lead to misalignment with broader societal values. As emphasized in [10], feedback often comes from a narrow pool of labelers, predominantly English-speaking individuals, which introduces inter-labeler disagreement and makes alignment across diverse contexts more challenging. This underscores the importance of developing mechanisms to ensure that feedback reflects the perspectives of a wider range of stakeholders.

Despite these challenges, RLHF remains a cost-effective approach for improving model behavior without requiring massive scaling of model size. Experiments such as those conducted in [19] have demonstrated that RLHF can effectively enhance alignment while maintaining computational efficiency. Models like InstructGPT have shown the ability to generalize to tasks outside their direct supervision, further illustrating RLHF's potential for scaling AI systems. However, limitations persist. Models trained with RLHF can still produce biased or harmful outputs, influenced by the biases present in their pretraining data. To address these risks, a multi-layered safety framework that combines RLHF with other mitigation techniques is essential for

ensuring safe and reliable AI behavior in real-world applications [21, 13].

The societal impacts of RLHF are equally significant. While RLHF holds promise for aligning AI systems with human preferences, it carries inherent risks, such as generating harmful or discriminatory content. Striking a balance between accessibility and control over AI technology is critical. Open-sourcing models trained with RLHF, for instance, could enable misuse in generating misinformation or hateful content. To mitigate these risks, it is crucial to implement robust safety measures, including monitoring for misuse and enforcing rate-limiting mechanisms. Additionally, as discussed in [10], ethical considerations such as transparency, accountability, and fairness must be carefully addressed to ensure that AI systems serve diverse societal contexts effectively.

In conclusion, RLHF provides a promising approach to improving the alignment of AI systems, particularly LLMs. However, its success depends on addressing both technical and ethical challenges. A holistic approach to AI alignment, integrating RLHF with comprehensive safety protocols and diverse feedback mechanisms, will be vital to ensuring that AI systems balance performance, fairness, and safety, ultimately serving the broader goals of society.

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